

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate Prospective Efficacy and Safety Data of Current FIX Prophylaxis Replacement Therapy in Adult Hemophilia B Subjects (FIX:C≤2%) or Current FVIII Prophylaxis Replacement Therapy in Adult Hemophilia A Subjects (FVIII:C≤1%) **Short Title/Acronym:** Phase 3 Lead-in Study for Hemophilia A and B **Protocol Number:** C0371004 **NCT Number:** NCT03587116 **EudraCT Number:** 2017-001271-23 **Sponsor Name:** Pfizer **Investigational Product:** Standard of Care FIX Replacement Therapy and Standard of Care FVIII Replacement Therapy **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To establish baseline prospective efficacy data (Annualized Bleeding Rate) of current FIX prophylaxis replacement therapy in the usual care setting of hemophilia B subjects, and current FVIII prophylaxis replacement therapy in hemophilia A subjects, prior to Phase 3 gene therapy studies. **Secondary Objectives:** To evaluate the annualized factor consumption while on usual care prophylaxis replacement therapy and to prospectively collect safety parameters.

2.2 Background & Rationale To establish a baseline and prospective efficacy/safety comparator of the standard of care for participants who will subsequently enroll in the Phase 3 gene therapy trials (AFFINE for giroctocogene fitelparvovec in Hemophilia A and BENEENE-2 for fidanacogene elaparpvovec in Hemophilia B). This prospective data collection is necessary to evaluate the non-inferiority and superiority of the subsequent one-time gene therapies in reducing bleeding episodes compared to routine replacement therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Prospective, Non-interventional (Lead-in trial) **Control Type:** Single-arm (Standard of care prophylaxis) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Hemophilia A cohort enrollment completed (\$N=101\$); Hemophilia B enrollment dependent on BENEENE-2 trial size requirements. **Number of Sites:** Multi-regional (18 countries across North America, South America, Asia

Pacific, Europe, and the Middle East). **Estimated Study Start:** Mid-2018 **Estimated Primary Completion:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males ≥ 18 and < 65 years of age with moderately severe to severe hemophilia A (documented FVIII activity $\leq 1\%$) or hemophilia B (documented FIX activity $\leq 2\%$) prior to baseline visit. 2. Previous experience with FVIII or FIX therapy (≥ 150 documented exposure days to a FVIII protein product, or ≥ 50 documented exposure days to a FIX protein product). 3. On a stable FVIII or FIX prophylaxis replacement therapy at study entry with the intention to remain on it for the duration of the study. 4. Evidence of a signed and dated informed consent document.

4.2 Exclusion Criteria 1. History of FVIII or FIX inhibitor (clinical or laboratory-based assessment). 2. Positive for hepatitis B surface antigen (HbsAg) or detectable HBV-DNA; or currently undergoing anti-viral therapy for chronic hepatitis C or testing positive for HCV-RNA. 3. Known history of human immunodeficiency virus (HIV) unless stable with adequate CD4 count ($>200/\text{mm}^3$) and undetectable viral load (<50 gc/mL). 4. Positive for neutralizing antibodies (nAb) to AAV6 (for Hemophilia A) or AAV-Spark100 (for Hemophilia B).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Usual care routine prophylaxis replacement therapy with FIX (for Hemophilia B) or FVIII (for Hemophilia A) according to standard clinical practice. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** At least 6 months of prospective data collection during the lead-in period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications for related co-morbidities as prescribed by the HCP. **Prohibited:** Nonfactor treatments and restricted therapies that confound efficacy data; any prior treatment with AAV gene-based therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent, Medical History, Viral serology (HIV, HBV, HCV), neutralizing antibodies (nAb) testing
Baseline	Day 0	Eligibility Confirmation, Start of Prospective Data Collection
Follow-up	Months 1 to ≥ 6	Prospective collection of bleeding rates, factor consumption, and safety parameters
End of Study		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized Bleeding Rate (ABR) for all bleeds (treated and untreated) during the prospective data collection period. **Measurement Method:** Calculated as the number of bleeds over the number of days from the baseline visit to the end of study multiplied by 365.25.

7.2 Secondary & Exploratory Endpoints * Annualized factor consumption (total factor replacement therapy consumption in IU and dose). * Incidence of Serious Adverse Events (SAEs) and standard safety profile of prophylaxis replacement therapy.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and SAEs during the prospective lead-in period via patient logs and site visits. **Serious Adverse Events (SAEs):** Reported per standard safety guidelines (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** External/Internal safety review committee evaluating the overall gene therapy program safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy Analysis Set (all enrolled subjects who received standard of care prophylaxis and contributed prospective data). **Sample Size Justification:** Sized to provide adequate baseline comparators for non-inferiority and superiority testing in the subsequent Phase 3 gene therapy trials (AFFINE and BENEGENE-2). **Handling of Missing Data:** ABR and factor consumption are annualized and calculated based on the available observed duration (extrapolated based on actual days on study).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine site monitoring for data quality regarding bleed records and factor consumption logs. **Audit Trail:** Standard clinical trial data management, electronic case report form (eCRF) locking, and source data verification.

11. Study Identifiers & Governance

Primary ID: C0371004 **Registry Number:** NCT03587116 **EudraCT Number:** 2017-001271-23 **Sponsor/Lead Organization:** Pfizer **Study Title:** Phase 3 Lead-in Study for Hemophilia A and B **Official Regulatory Title:** A Study to Evaluate Prospective Efficacy and Safety Data of Current FIX Prophylaxis Replacement Therapy in Adult Hemophilia B Subjects (FIX:C≤2%) or Current FVIII Prophylaxis Replacement Therapy in Adult Hemophilia A Subjects (FVIII:C≤1%)

12. Clinical Framework (Hemophilia Specific)

Primary Condition: Hemophilia A and Hemophilia B **Diagnosis Threshold:** FVIII:C $\leq 1\%$ (Hemophilia A) or FIX:C $\leq 2\%$ (Hemophilia B) **Medical Methodology:** Standard of Care Prophylaxis Replacement Therapy **Optimization Target:** Accurate baseline establishment of annualized bleeding rate (ABR) prior to gene therapy

13. Study Procedures & Methodology

Management Pathway: Routine prophylaxis replacement therapy in the usual care setting. **Education Protocol (HCP):** Training on bleeding rate capture and factor consumption logging. **Education Protocol (Patient):** Patient diary/log training for accurate bleeding episode and infusion recording. **Quality Audit Mechanism:** Source data verification of patient-reported bleeding events and factor utilization against prescription records.

14. Observation & Follow-Up Schedule

Total Duration: At least 6 months (approx. 24 weeks) of prospective follow-up **Onsite Visit Cadence:** Baseline and periodic follow-ups per standard of care and protocol schedules **Remote Monitoring Frequency:** Regular review of bleeding and infusion logs **Checklist Review Interval:** Review of patient diaries/logs at each follow-up contact

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	06-Apr-2026				
Clinical Operations Lead	[Name]	____	06-Apr-2026		Lead		
Statistician	[Name]	____	06-Apr-2026				